King Fahad Central Hospital

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CTA Protocol for Stroke Patients with Contrast Allergy

Balancing Time-Critical LVO Detection Against Contrast Reaction Risk — Aligned with AHA/ASA Guidelines

1. Guiding Principle

In suspected large vessel occlusion (LVO) stroke, CT angiography materially changes treatment eligibility and outcome. A history of contrast allergy — even significant — should prompt a rapid, structured risk-mitigation pathway, not an automatic cancellation of CTA. Rigid application of elective (multi-hour) premedication protocols is not appropriate in the time-critical acute stroke setting and may itself cause harm by delaying LVO detection and thrombectomy referral. Every decision on this pathway should be made jointly and quickly by the stroke team and radiologist, weighing the specific reaction history against the time-sensitivity of the clinical scenario.

2. Allergy History — Rapid Assessment

- ☐ Confirm the specific prior reaction (not just the label "allergy"): timing, symptoms, severity, treatment required, and whether it was to IV iodinated contrast specifically
- ☐ **Shellfish or topical iodine (e.g., Betadine) allergy alone is NOT a valid basis for withholding iodinated contrast — this is a common misconception; there is no clinically significant cross-reactivity between shellfish/iodine and iodinated contrast media, and this history alone does not require premedication or a change in plan**
- ☐ Distinguish a true prior contrast reaction from an unrelated IV site reaction, vasovagal episode, or expected sensation (warmth/metallic taste) — these do not require premedication
- ☐ Determine whether the prior reaction was mild (limited urticaria/itching), moderate (diffuse urticaria, mild bronchospasm, vomiting), or severe (anaphylaxis, laryngeal edema, hypotension, cardiovascular collapse)

3. Risk-Stratified Decision Pathway

Prior Reaction History	Recommended Approach
No known contrast reaction; shellfish/iodine "allergy" only	Proceed with standard CTA without delay; no premedication indicated
Mild prior reaction (e.g., limited hives/itching)	Proceed with CTA without delay in a time-critical LVO work-up; consider expedited premedication only if it will not meaningfully delay imaging — individualize with radiologist
Moderate prior reaction (diffuse urticaria, mild bronchospasm, vomiting)	Rapid joint discussion between stroke physician and radiologist; proceed with accelerated/emergent premedication regimen if feasible without unacceptable delay (see Section 4); if not feasible in time available, proceed with contrast and heightened monitoring/readiness to treat, per joint risk-benefit decision, OR consider non-contrast alternative pathway (Section 5) if clinical scenario allows
Severe prior reaction (anaphylaxis, laryngeal edema, hypotension/cardiovascular collapse)	Immediate joint discussion between stroke physician, radiologist, and (if available) allergy/anesthesia; strongly consider non-contrast alternative pathway (Section 5) if it will not unacceptably delay the treatment decision; if CTA still deemed necessary, proceed only with full resuscitation readiness at bedside and accelerated premedication if any time allows

These are default starting points for a rapid team discussion, not rigid rules — the treating stroke physician and radiologist make the final joint decision based on the specific patient and time available.

4. Premedication Regimens

Exact agents/doses/timing must be set by Radiology and Pharmacy per current departmental policy — the structure below is illustrative only. No premedication regimen eliminates reaction risk, including severe reactions; monitoring and readiness to treat remain essential regardless of premedication.

Regimen	Illustrative Structure
Standard elective premedication (NOT typically usable in acute stroke time window)	Corticosteroid (e.g., prednisone) at defined intervals over approximately 12–13 hours before contrast, plus an antihistamine before the study — generally impractical for emergent LVO work-up
Accelerated / emergent premedication (where departmental protocol supports it and some time is available)	IV corticosteroid (e.g., hydrocortisone or methylprednisolone) given immediately and, if time allows, repeated at a shorter interval before imaging, plus IV/IM antihistamine shortly before the study — per current departmental emergent-premedication protocol
No premedication feasible (true time-critical emergency)	Proceed per joint risk-benefit decision with full anaphylaxis-management readiness at bedside (Section 6); document that standard premedication was not feasible given the emergent LVO evaluation

5. Non-Contrast / Alternative Imaging Pathway

☐ Non-contrast CT plus MR angiography (time-of-flight, no contrast required) can substitute for CTA to assess for proximal LVO if MRI is rapidly available and will not unacceptably delay the treatment decision

☐ Note: CT perfusion for extended-window (6–24h) patient selection also requires iodinated contrast — the same risk-stratified decision applies; MR perfusion/diffusion imaging is an alternative if MRI access and timing allow

☐ Ultrasound (carotid/transcranial Doppler) does not reliably substitute for CTA/MRA in acute LVO localization and is not a recommended primary alternative in this time-critical setting

☐ If no contrast-free alternative is feasible within an acceptable time frame, proceed with the risk-stratified contrast pathway in Sections 3–4 rather than forgoing LVO assessment entirely

6. Administration Precautions & Emergency Readiness

☐ Use the lowest effective contrast volume and, where available per departmental protocol, a low-osmolar or iso-osmolar non-ionic contrast agent for patients with any allergy history

☐ Anaphylaxis/emergency medications (epinephrine, antihistamine, corticosteroid) and airway equipment immediately available at the CT suite for any patient with a moderate or severe reaction history

☐ Physician or advanced practice provider readily available (in-suite or immediately adjacent) during contrast injection for moderate/severe-history patients

☐ Continuous observation during injection; extended observation period after injection for higher-risk patients per departmental protocol

☐ Technologist/nurse briefed on allergy history and reaction plan before injection begins

7. Communication & Documentation

☐ Allergy history, severity classification, and the joint stroke-physician/radiologist decision documented in the chart before contrast administration

☐ Premedication given (or reason premedication was not feasible) documented with times

☐ Any reaction occurring during or after the study documented, including treatment given, and reported per institutional adverse-event/pharmacovigilance policy

☐ Allergy and reaction history updated in the patient's permanent allergy record for future encounters

Radiology / Stroke Team Sign-Off

Stroke Physician: ____________________________ Date/Time: _______________

Radiologist: ________________________________ Decision: Proceed with contrast / Premedicate / Non-contrast alternative

References: 2026 AHA/ASA Guideline for the Early Management of Patients With Acute Ischemic Stroke and subsequent focused updates; American College of Radiology (ACR) Manual on Contrast Media premedication guidance; institutional Radiology contrast-reaction and emergent-premedication protocols. This template must be re-verified against current departmental contrast policy, current ACR guidance, and the current AHA/ASA guideline version at the time of institutional adoption.